Unit1: Linear Systems and Structural Foundations

1. For each of the following systems of linear equations, draw the row picture and the column picture. Hence, determine the solution of the system.

   $a)\ x - y = 2,\quad 2x + y = 7$ $\qquad Ans: (3,1)$

   b) $x - y = -1,\ 2x + y = 4\ \ (HW)$ $\quad Ans: (1,2)$

   c) $2x + y = 5,\quad x - y = 7$ $\qquad$ Ans: (4, -3)

   d) $x + 2y = 2,\quad 2x + 4y = 6$ $\qquad Ans: No\ solution$

   e) $x + 2y = 3,\quad 2x + 4y = 6\ \ (HW)$ $Ans: Many\ solutions$

2. Find the rank of the following matrices:

$$\text{i) } A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 5 & 4 \\ 2 & 4 & 2 \end{bmatrix} \text{ ii) } A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 3 & 5 & 4 \\ 4 & 8 & 4 \end{bmatrix} \text{ iii) } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 1 & 2 \end{bmatrix}$$

Ans: $\text{i) } \rho(A) = 2 \text{ ii) } \rho(A) = 2 \text{ ii) } \rho(A) = 3$

3. Reduce the following matrices to Row Reduced Echelon form:

$$\text{i) } A = \begin{bmatrix} 1 & 1 & 2 & 3 \\ 2 & 3 & 5 & 8 \\ 1 & 2 & 3 & 5 \end{bmatrix} \qquad \text{ii) } A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{bmatrix}$$

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4. Test for consistence and solve using Gauss elimination method

   a) $x + 2y - z = 6,\ 2x + y + z = 3, x - y + z = -2.$ **Ans: (1, 2, -1)**

   b) $x - 3y - 8z + 10 = 0,\ 3x + y - 4z = 0, 2x + 5y + 6z = 13.$ **Ans: (2k-1, -2k+3, k)**

   b) $x + 2y - 4z = -4,\ 2x + 5y - 9z = -10,\ y - z = 2.$ **Ans: No solution**

   Home work problems: (a and b)

   a) $2x + y - z = 4,\ x + y + z = 4,\ x - y + z = 2.$ $\qquad$ Ans: (2, 1, 1)

b) $x + 2y - 3z = 1,\ y - 2z = 2,\ 2y - 4z = 4$. Ans: (-3-K, 2+2K, K)

5. These equations are certain to have a solution $x = y = 0$. For which values of a is there a whole line of solutions? $ax + 2y = 0, 2x + ay = 0$.

Ans: $a = \pm 2$.

6. Find the values of a and b for the system $x + y + z = 5, ax + y + az = 3, (1 + a)x + 2y + 3z = b$ to possess a) a unique solution b) infinitely many solutions c) no solution.

Ans: $a)\quad a \neq 2,\ b\ \in R\ (any\, real\, number)\quad b)\ a = 2,\ b = 8,\quad c)\ a = 2,\ b \neq 8.$

7. Determine the values of a and b for the system $x + y + az = 2b,\ x + 3y + (2 + 2a)z = 7b,\ \ 3x + y + (3 + 3a)z = 11b$ will have a) trivial solution b) a unique non-trivial solution c) no solution, d) infinitely many solutions. (HW)

Ans: $a)\quad a \neq -5,\ b = 0\quad b)\ a \neq -5\quad c)\ a = -5,\ b \neq 4/9\quad d)\ a = -5,\ b = 4/9.$

H.W.

Consider the system of equations $x + 2y + z = 3, ax + 5y = 10, 2x + 7y + az = b$.

a) Find the values of a for which the system has a unique solution.
b) Find the pair of values (a, b) for which the system has infinitely number of solutions.

Ans: $a)\ a \neq 5,\ and\ a\ \neq -3,\quad b)\ (5,12), (-3, -4).$

8. Which three matrices $E_{21}, E_{31},\ E_{32}$ put A into triangular form U?

$A = [1\ 1\ 0\ 4\ 6\ 1\ \ -2\ 2\ 0\ ]$

9. Find LU and LDU factorization for

i) $A=\begin{bmatrix} 3 & 1 & 2 \\ 2 & -3 & -1 \\ 1 & 2 & 1 \end{bmatrix}$, ii) $A=\begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 4 & 2 & 1 \\ 1 & 3 & 2 & 1 \\ 1 & 3 & 4 & 1 \end{bmatrix}$ (HW), iii) $A=\begin{bmatrix} 1 & 2 & 0 & 2 \\ 1 & 3 & 2 & -1 \\ 2 & 3 & 4 & 0 \end{bmatrix}$

10. Find the inverse of the following matrices using Gauss-Jordon method

ii) $A=\begin{bmatrix} 2 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & 2 \end{bmatrix}$ ii) $A=\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ (HW)

Ans: i) $\begin{bmatrix} -1 & 1 & 1 \\ 5/3 & -1 & -4/3 \\ -1/3 & 0 & 2/3 \end{bmatrix}$ ii) $\begin{bmatrix} 3/4 & 1/2 & 1/4 \\ 1/2 & 1 & 1/2 \\ 1/4 & 1/2 & 3/4 \end{bmatrix}$

11. Which of the following sets, under standard addition and scalar multiplication, is NOT a vector space?

A) The set of all polynomials of degree exactly 2

B) The set of all polynomials of degree at most 2

C) The set of all continuous functions on [0, 1]

D) The set of all 2×2 matrices

Answer: Option A

Explanation: A vector space must be closed under addition and scalar multiplication and must contain the zero vector. The set of polynomials of degree exactly 2 is NOT a vector space because the sum of two degree-2 polynomials can cancel leading terms (e.g., $x^2 + (-x^2) = 0$), giving degree less than 2 — violating closure. It also doesn't contain the zero polynomial. Options B, C, D all satisfy all vector space axioms.

Q.12. Let W be a subset of a vector space V. Which condition is SUFFICIENT to guarantee that W is a subspace of V?

A) W is non-empty and closed under addition only

B) W is non-empty and closed under scalar multiplication only

C) W contains the zero vector and is closed under addition and scalar multiplication

D) W has the same dimension as V

Answer: Option C

Explanation: The Subspace Test requires three conditions: (1) the zero vector is in W, (2) W is closed under addition, and (3) W is closed under scalar multiplication. Non-emptiness combined with conditions (2) and (3) also guarantees the zero vector is in W (take $0 \cdot w$). Option A fails without scalar closure, Option B fails without additive closure, and Option D is irrelevant — a proper subspace can have lower dimension than V.

Q.13. Consider the set $S = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$. What is S?

A) A line through the origin in $\mathbb{R}^3$

B) A plane through the origin in $\mathbb{R}^3$, hence a subspace

C) A plane not through the origin, hence not a subspace

D) All of $\mathbb{R}^3$

Answer: Option B

Explanation: S is defined by a single homogeneous linear equation $x + 2y - z = 0$, which geometrically defines a plane through the origin in $\mathbb{R}^3$. It passes through the origin (0,0,0 satisfies it), it's closed under addition and scalar multiplication (both preserve the equation). This makes S a 2-dimensional subspace of $\mathbb{R}^3$. A non-homogeneous equation like $x + 2y - z = 5$ would give a plane NOT through the origin, which is not a subspace.

Q.14. If W is a subspace of $\mathbb{R}^4$ with dim(W) = 4, then W must be:

A) A proper subset of $\mathbb{R}^4$

B) Equal to $\mathbb{R}^4$ itself

C) The null space of some 4×4 matrix

D) Spanned by exactly 4 orthogonal vectors

Answer: Option B

Explanation: A subspace of $\mathbb{R}^n$ with dimension n must be all of $\mathbb{R}^n$ itself. Since $\mathbb{R}^4$ has dimension 4 and W $\subseteq \mathbb{R}^4$ with dim(W) = 4, W must contain 4 linearly independent vectors, which form a basis for $\mathbb{R}^4$, so W = $\mathbb{R}^4$. This is a key dimensional argument: a subspace can only equal the entire space when their dimensions match.

Q.15. **Determine whether the vectors v1 = (1,2,1), v2 = (2,1,0), v3 = (0,3,2) are linearly independent.**

Answer one can verify: v3 = 2v1 - v2: 2(1,2,1) - (2,1,0) = (0,3,2) = v3

Q.16.**Determine whether the vectors v1 = (1,3,2), v2 = (2,1,4), v3 = (3,2,6) are linearly dependent.**

Ans :dependent

Q.17. **Determine whether the vectors v1 = (1,-3,2), v2 = (2,1,-3), v3 = (-3,2,1) are linearly independent and hence find the relationship between them.**

Ans Dependent v1+v2+v3=0**.**

Q.18. Show that the set of vectors {u+v,u-v, u-2v+w} is linearly independent. (HW)

Q.19. **Do these vectors {(2,2,3), (-1,-2,1), (0,1,0)} span** R^3.

Ans: yes. They span R^3.

20. Find the basis and dimension for each of the four subspaces of the following matrices and verify the Rank-nullity theorem:

$$\text{i) } A=\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}, \text{ ii) } A=\begin{bmatrix} 4 & 2 & 4 \\ 2 & 4 & 8 \\ 2 & 2 & 2 \end{bmatrix}$$

Homework problems

$$\text{i) } A=\begin{bmatrix} 1 & 2 & 3 & 4 \\ 3 & 7 & 10 & 11 \\ 2 & 4 & 7 & 7 \end{bmatrix}, \text{ ii) } A=\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 2 & 1 \end{bmatrix}$$

Q.21.Find the basis of the plane 2x - y + z -3t = 0.

Ans: {(1/2, 1, 0, 0), (-1/2, 0, 1, 0) and (3/2, 0, 0, 1)}

Q.22 Which of these transformations are not Linear? Give reasons

a) T(x,y,z) =(x + y +z , 2x - 3y + 4z)

b) T(x,y) =(x + 3 , 2y, x+y)

c) T(x,y)=(xy,x)

Ans: b) T(0) is not equal to 0

d) T(kv)$\neq kT(v)$

Q. 23. Which of these transformations are not Linear? Give reasons

a) $x+2y+z-3=0$

b) $x+5y-2z=2t$

c) $x+2y+z=0$

d) $-5x-y-3z=0$

Ans: a) Because T (0) not equal to zero

24. Find the matrix of the differentiation operator $\left(A_{diff}\right)$ on the vector space of polynomials of degree at most 4 with respect to the standard basis.

Ans: $$A_{diff}=\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

25. Find the matrix of the integration operator on the vector space of polynomials of degree at most 3 with respect to the standard basis. (HW)

Ans: $$A_{Int} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 1/4 \end{bmatrix}$$

26. From the vector space of cubic polynomials P_3 to the fourth-degree polynomials P_4, find the matrix representing multiplication by $3t+5$?

Ans:

$$A = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

27. Find the matrix of the linear transformation $T:\mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x,y,z) = (2x-y,\ 3y+z)$ with respect to the standard bases $\mathbb{R}^3 : \{(1,0,0),(0,1,0),(0,0,1)\}$ and $\mathbb{R}^2 : \{(1,0),(0,1)\}$. And also find the Range and kernel of T.

Ans: $T = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$; Range = $\mathbb{R}^2$, Kernel = line in $\mathbb{R}^3$.

28. Find the matrix of the linear transformation T on $\mathbb{R}^3$ defined by $T(x,y,z) = (2y+z, x-4y, 3x)$ with respect to the standard basis $(1,0,0),(0,1,0),(0,0,1)$. Also find the Range and kernel of T (HW).

Ans: $T = \begin{bmatrix} 0 & 2 & 1 \\ 1 & -4 & 0 \\ 3 & 0 & 0 \end{bmatrix}$, Range is the $\mathbb{R}^3$, kernel is the origin the $\mathbb{R}^3 0$

29. Find the matrix that rotates every vector in $\mathbb{R}^2$ through 90° in the positive sense and then projects the result onto the x axis.

Ans: $Q = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, A = PQ = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$

30. Find the matrix that first projects every vector in R^2 onto the x-axis and then rotates it by 90°. (HW)

Ans: $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, Q = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, A = QP = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

31. Find the matrix that reflects every vector in R^2 about the y-axis. Hence, find the image of the vector $\begin{bmatrix} 4 \\ -3 \end{bmatrix}$.

Ans:

$$H = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, H\begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$$